

Characterizing second generation biofuels: Excess enthalpies and vapour-liquid equilibria of the binary mixtures containing 1-pentanol or 2-pentanol and n-hexane



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ABSTRACT

Experimental data of excess enthalpies and vapour-liquid equilibria (VLE) are reported for the binary systems (1-pentanol + n-hexane) and (2-pentanol + n-hexane). An isothermal flow calorimeter was used to determine excess enthalpies at $T = (298.15 \text{ and } 313.15) \text{ K}$ and the results were fitted using Redlich-Kister equation. Vapour-liquid equilibria data were measured using an isothermal total pressure cell at $T = 313.15 \text{ K}$. Barker's method allows to correlate these data using Margules, Wilson, NRTL and UNIQUAC equations. An endothermic behavior is observed when 1-pentanol or 2-pentanol is mixed with n-hexane whose effect increases at higher temperatures and both systems present a positive deviation from the Raoult's law.

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1. Introduction

Energy policies in the European Union focus on the promotion of the use of biofuels, as is established in Directive «20-20-20» [1], which determines a minimum content of 10% of biofuels overall gasoline and diesel fuels.

Properties of these new biofuels change in comparison with fossil fuels, therefore a characterization of these properties is required in order to enhance their use. In this sense, our research group is involved in the measurement of thermodynamic and thermophysical properties, such as density, viscosity, heat capacity, excess enthalpy or vapour-liquid equilibria, of mixtures of interest for the formulation of biofuels.

In this paper, 1-pentanol and 2-pentanol are studied as oxygenated additives of renewable origin due to their qualities such as high octane rating or high combustion heat, and n-hexane was chosen as hydrocarbon. Excess enthalpies and vapour-liquid

equilibria of the mixtures (1-pentanol + n-hexane) and (2-pentanol + n-hexane) are reported.

2. Experimental section

2.1. Materials

1-Pentanol, 2-pentanol and n-hexane were chromatography quality reagents of the highest purity available from Sigma-Aldrich. Their purities were $>0.997 \text{ (GC)}$, $>0.98 \text{ (GC)}$ and $>0.99 \text{ (GC)}$, respectively and were also checked by gas chromatography being better than 0.997. The details are summarized in Table 1.

2.2. Experimental techniques

Both techniques, used in this work, were described in detail in previous papers [2,3] and a brief summary of their main characteristics is reported below.

A quasi-isothermal flow calorimeter, developed in our laboratory, was used to determine excess enthalpies [2]. The technique is based on the accurate measurement of the energy required to maintain the mixing vessel at a constant temperature, using a Peltier cooler to remove a constant heat rate from the flow cell, and

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Table 1

Material description.

Chemical name	Source	Mass fraction purity (GC) ^a	Purification method
1-Pentanol	Sigma-Aldrich	≥0.997	None
2-Pentanol	Sigma-Aldrich	≥0.997	None
n-Hexane	Sigma-Aldrich	≥0.997	None

^a From the supplier.

a control-heater to compensate this energy and the energy liberated (exothermic mixing) or absorbed (endothermic mixing) by the mixing process. The properties measured directly and their corresponding standard uncertainties are: flow rate $\pm 0.15\%$ (≤ 10 mL/min); temperature ± 10 mK; the heating and cooling systems have a stability better than $2.4 \mu\text{W}$. The estimated expanded uncertainties, for a cover factor $k = 2$, are: ± 0.0002 in the composition and $\pm 1\%$ in the excess molar enthalpy.

Vapour-liquid equilibrium measurements were carried out with an isothermal total pressure cell [3]. The equilibrium properties measured directly and their standard uncertainties are: injected volume ± 0.03 mL, temperature ± 10 mK and total pressure ± 5 Pa. The overall mole fraction is determined accurately from the volumetric displacement of the injector pistons, the temperature of the injectors and the densities for the pure components, obtaining an estimated standard uncertainty for composition of ± 0.0005 . Experimental values were obtained in two overlapping runs starting from opposite ends of the composition range and the repeated central compositions are used for checking the quality of the measurements.

3. Results

Excess enthalpies were measured at $T = (298.15 \text{ and } 313.15) \text{ K}$ for the binary mixtures (1-pentanol + n-hexane) and (2-pentanol + n-hexane). These data, reported in Table 2 and in Fig. 1, were correlated using a modified Redlich-Kister [4] equation:

$$\frac{H_m^E}{\text{J} \cdot \text{mol}^{-1}} = x(1-x) \sum_{i=1}^n \frac{A_i(2x-1)^{i-1}}{[1+C(2x-1)]} \quad (1)$$

The parameters of equation (1) and the standard deviations are given in Table 3. The optimal number of parameters was selected by examining F-test [5].

As regards the use of a static technique for measuring VLE, the vapour phase need not be sampled for analysis and the data are thermodynamically consistent “per se” [6]. Barker's method [7,8] was applied for data reduction and the virial equation of state was used for modelling vapour phase non-ideality. Second virial coefficients (B_{ii} , B_{ij}) were calculated by the Hayden and O'Connell method [9] using the parameters given by Dymond and Smith [10]. The average values of the experimental vapour pressures (p_i^{sat}) for the pure compounds and their comparison with the literature [11–22], the molar volumes of pure liquids (V_i^{L}) and the second virial coefficients (B_{ii} , B_{ij}) are indicated in Table 4.

Margules equation up to six-parameter [23], Wilson [24], NRTL [25] and UNIQUAC [26] models were applied for correlating the experimental data. The expressions are given by Eqs. (2)–(5), respectively:

$$\frac{G_m^E}{RT} = [A_{ji}x_i + A_{ij}x_j - (\lambda_{ji}x_i + \lambda_{ij}x_j)x_i x_j + (\eta_{ji}x_i + \eta_{ij}x_j)x_i^2 x_j^2] x_i x_j \quad (2)$$

Table 2Experimental excess molar enthalpies H_m^E as a function of the mole fraction x_1 for the binary systems^a at atmospheric pressure.

x_1	$H_m^E/\text{J} \cdot \text{mol}^{-1}$	x_1	$H_m^E/\text{J} \cdot \text{mol}^{-1}$	x_1	$H_m^E/\text{J} \cdot \text{mol}^{-1}$
1-Pentanol (1) + n-hexane (2) at $T = 298.15 \text{ K}$					
0.0000		0.3522		0.7015	
	0.0		537.8		290.1
0.0504		0.4030		0.7514	
	329.3		524.2		235.6
0.0999		0.4528		0.8021	
	413.2		505.8		178.8
0.1509		0.5035		0.8518	
	465.6		477.6		126.6
0.2010		0.5535		0.9004	
	501.3		441.6		69.6
0.2524		0.6025		0.9499	
	526.9		397.1		21.0
0.3027		0.6525		1.0000	
	533.6		346.3		0.0
1-Pentanol (1) + n-hexane (2) at $T = 313.15 \text{ K}$					
0.0000		0.3503		0.7016	
	0.0		724.5		415.4
0.0504		0.4032		0.7515	
	455.1		711.6		340.1
0.1000		0.4530		0.8019	
	569.2		688.6		264.8
0.1510		0.5039		0.8517	
	637.7		652.0		200.0
0.2011		0.5539		0.9004	
	680.1		605.1		128.2
0.2525		0.6027		0.9499	
	705.9		547.3		67.4
0.3007		0.6527		1.0000	
	720.9		487.4		0.0
2-Pentanol (1) + n-hexane (2) at $T = 298.15 \text{ K}$					
0.0000	0.0	0.3527	834.8	0.7018	637.0
0.0500	442.7	0.4013	847.5	0.7519	546.9
0.1016	575.7	0.4530	848.7	0.7992	449.7
0.1523	663.0	0.5018	835.9	0.8509	341.3
0.2021	724.9	0.5518	813.0	0.8999	229.2
0.2509	770.3	0.6027	769.7	0.9496	108.9
0.3012	808.2	0.6528	709.8	1.0000	0.0
2-Pentanol (1) + n-hexane (2) at $T = 313.15 \text{ K}$					
0.0000	0.0	0.3530	1027.2	0.7019	774.8
0.0501	559.9	0.4015	1032.7	0.7519	670.0
0.1017	740.8	0.4533	1028.7	0.7992	562.8
0.1524	851.9	0.5022	1006.4	0.8509	427.9
0.2022	923.7	0.5521	977.5	0.9016	284.5
0.2512	970.7	0.6031	928.2	0.9513	133.1
0.3014	999.3	0.6531	855.0	1.0000	0.0

^a Standard uncertainties ($k = 1$): $u(x_1) = 0.0001$, $u(T) = 10 \text{ mK}$, $u_r(H_m^E) = 0.5\%$.

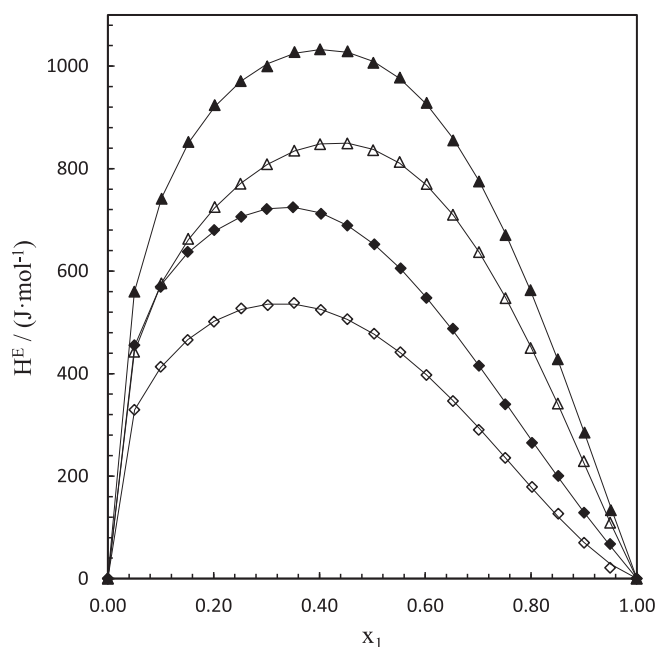


Fig. 1. Experimental excess molar enthalpies as a function of the composition for the binary systems: 1-pentanol (1) + n-hexane (2) (◇) at $T = 298.15$ K and (◆) at $T = 313.15$ K; 2-pentanol (1) + n-hexane (2) (Δ) at $T = 298.15$ K and (▲) at $T = 313.15$ K. The line represents the values calculated by Redlich–Kister equation.

Table 3

Parameters of Redlich–Kister Eq. (1), standard deviation of excess molar enthalpies σ and, absolute maximum deviation ($|\Delta H_{m,\max}^E|$) for the measured systems.

T (K)	C	A_1	A_2	A_3	A_4	A_5	σ	$ \Delta H_{m,\max}^E $
(J·mol ⁻¹)								
1-Pentanol (1) + n-hexane (2)								
298.15	0.97723	1916.52	589.28	−973.40	−530.06	−229.09	3.1	7.4
313.15	0.93272	2619.99	771.12	−1194.8	−320.09	516.03	2.6	4.2
2-Pentanol (1) + n-hexane (2)								
298.15	0.95900	3352.83	2462.85	−289.95	−974.04	−513.77	2.0	4.0
313.15	0.95612	4040.02	2856.45	132.07	−858.56	−853.80	2.9	5.2

$$\frac{G_m^E}{RT} = - \sum_i x_i \ln \left(\sum_j x_j A_{ij} \right) \quad (3)$$

$$\frac{G_m^E}{RT} = \sum_i x_i \left(\frac{\sum_j A_{ji} G_{ji} x_j}{\sum_k G_{ki} x_k} \right) \quad (4)$$

$$\frac{G_m^E}{RT} = \sum_i x_i \ln \left(\frac{\varphi_i}{x_i} \right) + \left(\frac{z}{2} \right) \sum_i q_i x_i \ln \left(\frac{\vartheta_i}{q_i} \right) - \sum_i q_i x_i \ln \left(\sum_j \vartheta_j A_{ji} \right) \quad (5)$$

where A_{ij} , A_{ji} and α (only for NRTL model) are the adjustable parameters, and $G_{ji} = \exp(-\alpha_{ji} A_{ji})$; $\vartheta_i = q_i x_i / \sum_j q_j x_j$; $\varphi_i = r_i x_i / \sum_j r_j x_j$ and $z = 10$.

Table 5 contains the experimental values of total pressure, liquid phase composition and vapour phase composition, calculated by Margules equation, for the binary systems (1-pentanol or 2-pentanol and n-hexane at $T = 313.15$ K, and both sets of data

Table 4

Average values of the experimental vapour pressures (P_i^s) for the pure compounds and literature values ($P_i^s(\text{lit.})$), the molar volumes of pure liquids (V_i^L) and the second virial coefficients (B_{ii} , B_{ij}) at $T = 313.15$ K used for the calculations.

	1-Pentanol ($i = 1$)	2-Pentanol ($i = 2$)	n-Hexane ($i = 3$)
P_i^s /(kPa)	0.905 ^{a,b}	2.295 ^{a,b}	37.188 ^a
$P_i^s(\text{lit.})$ /(kPa)	0.918 ^c	2.260 ^e	37.04 ⁱ
	0.872 ^d	2.281 ^f	37.239 ^j
	0.891 ^e	2.298 ^g	37.267 ^k
		2.280 ^h	37.278 ^l
		2.13 ⁱ	37.283 ^m
V_i^L /(cm ³ ·mol ⁻¹) ⁿ	110.1	111.2	134.5
B_{ii} /(cm ³ ·mol ⁻¹) ^o	-3001	-2560	-1649
B_{ij} /(cm ³ ·mol ⁻¹) ^o	-1973	-1839	-1649

^a Standard uncertainties ($k = 1$): $u(P) = 5$ Pa; $u(T) = 10$ mK.

^b Average of 7 runs.

^c Ref. [11].

^d Ref. [12].

^e Ref. [13].

^f Ref. [14].

^g Ref. [15].

^h Ref. [16].

ⁱ Ref. [17].

^j Ref. [18].

^k Ref. [19].

^l Ref. [20].

^m Calculated from Antoine equation using constants reported by Reid et al. [21].

ⁿ Ref. [22].

^o Calculated by Hayden et al. [9] from Dymond et al. [10].

have been represented in Fig. 2.

Finally, VLE correlation results are summarized in Table 6 which contains the values of the dimensionless adjustable parameters for the different models, the root mean square of the difference between experimental and calculated pressures (rms ΔP) and the maximum value of this difference (max $|\Delta P|$).

4. Discussion and conclusions

The study of the energy behavior of these mixtures shows that both systems have a strong endothermic effect due to the mixing process which increases with temperature. The highest excess enthalpies are 724 J mol⁻¹ for the system with 1-pentanol at an alcohol mole fraction of 0.35 and 1033 J mol⁻¹ for a mole fraction of 2-pentanol of 0.40, both at 313.15 K. The maximum values decrease to 528 J mol⁻¹ and 849 J mol⁻¹ at 298.15 K and mole fractions of 0.35 and 0.45 for 1-pentanol and 2-pentanol, respectively, as can be observed in Fig. 1. The higher endothermic behavior of the mixtures with 2-pentanol in comparison with 1-pentanol was also observed in previous studies [27–31].

The four systems are well correlated using modified Redlich–Kister equation with six parameters, and the standard deviations range from 2 to 3.1 J mol⁻¹. We have only found data for the system

Table 5

Total pressure VLE data, liquid mole fraction and vapour mole fraction calculated using Margules equation for the binary systems at $T = 313.15$ K.^a

x_1	$y_{1,calc}$	P (kPa)	x_1	$y_{1,calc}$	P (kPa)
1-pentanol (1) + n-hexane (2)					
0.0000	0.0000	37.188	0.5018	0.0204	31.426
0.0287	0.0077	36.627	0.5468	0.0215	30.411
0.0813	0.0120	36.210	0.5521	0.0217	30.404
0.1338	0.0134	35.774	0.5977	0.0231	29.178
0.1857	0.0143	35.310	0.6029	0.0233	29.141
0.2380	0.0152	34.807	0.6526	0.0253	27.617
0.2899	0.0162	34.271	0.7022	0.0281	25.752
0.3416	0.0172	33.692	0.7526	0.0320	23.409
0.3931	0.0182	33.040	0.8024	0.0378	20.537
0.4013	0.0183	33.041	0.8524	0.0473	16.984
0.4445	0.0192	32.295	0.9018	0.0659	12.652
0.4516	0.0193	32.289	0.9502	0.1172	7.474
0.4957	0.0202	31.431	1.0000	1.0000	0.920
2-pentanol (1) + n-hexane (2)					
0.0000	0.0000	37.188	0.5016	0.0515	30.590
0.0497	0.0204	36.773	0.5515	0.0554	29.432
0.1005	0.0268	36.315	0.5518	0.0554	29.459
0.1504	0.0302	35.805	0.6013	0.0599	28.106
0.2008	0.0331	35.258	0.6019	0.0600	28.153
0.2510	0.0360	34.647	0.6520	0.0657	26.651
0.3012	0.0389	33.993	0.7020	0.0729	24.866
0.3513	0.0419	33.273	0.7517	0.0826	22.750
0.4010	0.0449	32.480	0.8018	0.0964	20.176
0.4012	0.0449	32.453	0.8520	0.1185	17.003
0.4514	0.0481	31.557	0.9017	0.1596	13.108
0.4514	0.0481	31.603	0.9496	0.2594	8.456
0.5014	0.0515	30.574	1.0000	1.0000	2.307

^a Standard uncertainties ($k = 1$): $u(x_1) = 0.0005$, $u(y_{1,calc}) = 0.0005$, $u(P) = 5$ Pa; $u(T) = 10$ mK.

Table 6

VLE correlation results for the binary systems at $T = 313.15$ K: dimensionless adjustable parameters, root mean square pressure deviation (rms ΔP) and the maximum value of the deviation ($\max|\Delta P|$).

	Margules	Wilson	NRTL	UNIQUAC
1-Pentanol (1) + n-hexane (2)				
A_{12}	2.8030	0.0948	0.7829	1.7049
A_{21}	1.3554	0.6109	2.0893	0.2106
λ_{12}	4.7153			
λ_{21}	0.6402			
η_{12}	6.3642			
η_{21}	0.9037			
α_{12}			0.5397	
rms ΔP /kPa	0.070	0.131	0.194	0.452
$\max \Delta P $ /kPa	0.137	0.320	0.442	0.954
2-Pentanol (1) + n-hexane (2)				
A_{12}	2.5114	0.1473	0.8345	1.7023
A_{21}	1.3081	0.6580	1.8717	0.2389
λ_{12}	4.2414			
λ_{21}	0.6165			
η_{12}	5.4526			
η_{21}	0.6091			
α_{12}			0.6326	
rms ΔP /kPa	0.045	0.290	0.142	0.500
$\max \Delta P $ /kPa	0.088	0.540	0.310	0.926

(1-pentanol + n-hexane) at 298.15 K for comparison, which shown some discrepancies between them and with our results. The average absolute deviations are 29 J mol⁻¹ from [32], 14 J mol⁻¹ from [33], 20 J mol⁻¹ from [34] and 12 J mol⁻¹ from [35].

In addition, the best VLE data correlation is obtained using six-parameter Margules equation with a root mean square pressure residual of 70 Pa and a maximum deviation of 137 Pa for the system (1-pentanol + n-hexane), and 45 Pa and 88 Pa, respectively for the system (2-pentanol + n-hexane). In both systems, the worst fit is obtained with UNIQUAC model, Wilson adjustment is better than NRTL for the system with 1-pentanol and the opposite for the system with 2-pentanol. Pressure versus composition of both phases behavior is plotted in Fig. 1 where is shown that both systems exhibit a positive deviation from the ideality but no azeotrope is found. We have found VLE data for the system 2-pentanol + n-hexane at the same conditions [17]. Applying our reduction method to these data are obtained higher root mean square deviations: 0.372 with six-parameter Margules equation, 0.453 Pa with Wilson and 0.422 Pa with NRTL. The differences in the pure vapour pressures are also shown in Table 4, where these values are lower than ours and those reported in the literature. There are isobaric data for the system 1-pentanol + hexane [36,37] and isothermal data at different temperatures [38,39].

Finally, Figs. 3 and 4 represent the calculated excess functions of the binary systems at 313.15 K, 1-pentanol + hexane and 2-pentanol + hexane, respectively. The excess Gibbs energies were calculated using the Margules equation, the excess enthalpies were fitted with the Redlich-Kister equation, and finally the excess entropies were obtained using the relation between them: $G^E = H^E - T \cdot S^E$. For the system 1-pentanol + hexane, the excess Gibbs energy is higher than the excess enthalpy giving as a result an excess entropy negative in most of the composition range (from $x = 0.1$ to 1 for 1-pentanol), the maximum $G^E = 1082$ J mol⁻¹ at $x_1 = 0.45$. However, the excess enthalpy is slightly higher than the excess Gibbs energy obtaining an excess entropy positive in most of the composition range (from $x = 0$ to 0.75 for 2-pentanol) for the system 2-pentanol + hexane, with a maximum $G^E = 982$ J mol⁻¹ at $x_1 = 0.45$.

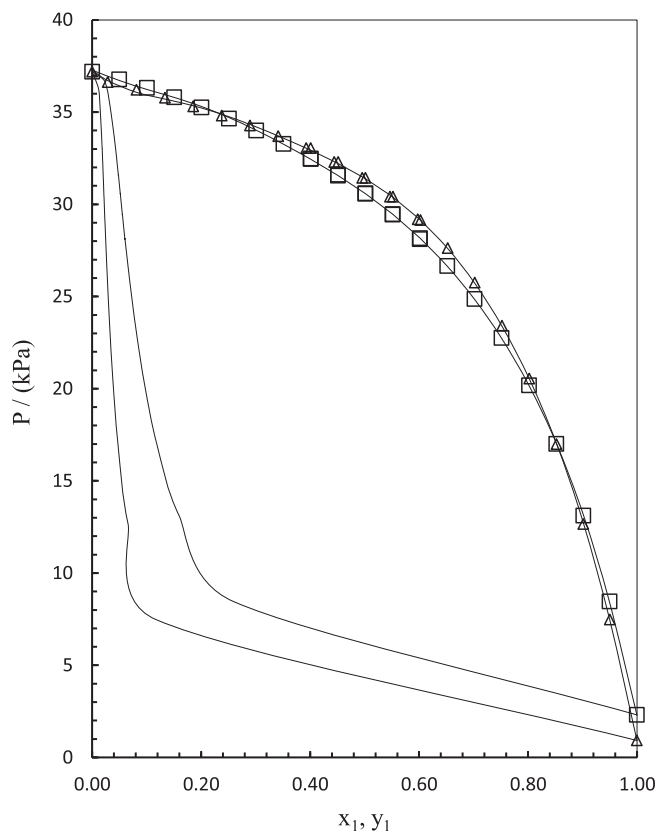


Fig. 2. Experimental VLE data at $T = 313.15$ K for the systems 1-pentanol (1) + n-hexane (2) (Δ) and 2-pentanol (1) + n-hexane (2) ($\square$). Calculated values using Margules equation with the parameters given in Table 6 (—).

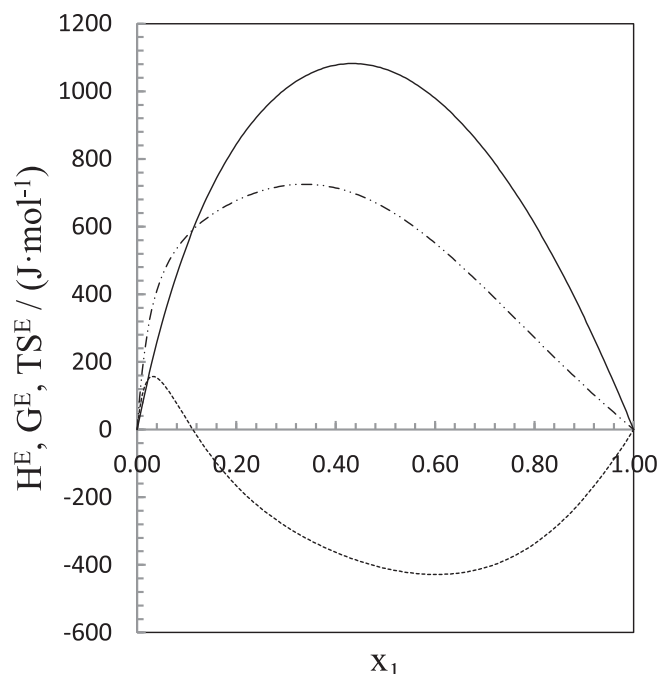


Fig. 3. Excess functions for the system 1-pentanol (1) + n-hexane (2) at $T = 313.15$ K: (—) G_m^E ; (---) H_m^E ; (- · - ·) TS_m^E . G_m^E was calculated using the Margules equation and H_m^E with the Redlich-Kister equation.

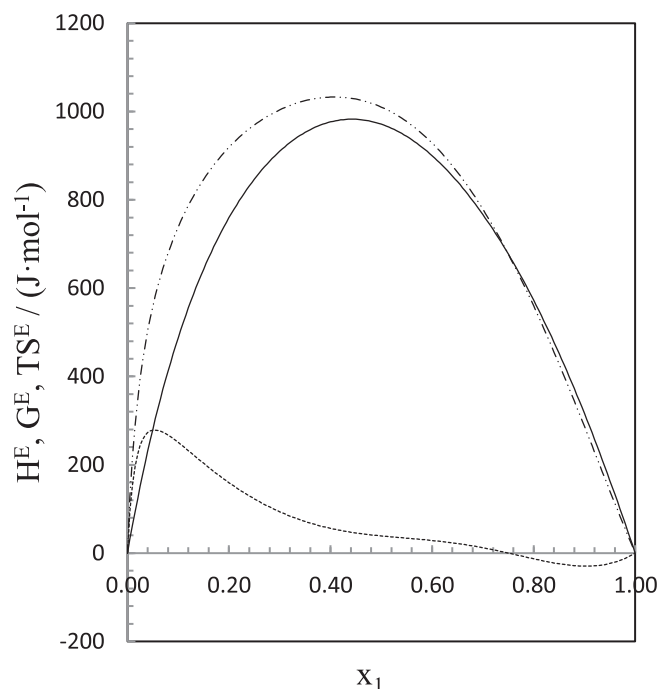


Fig. 4. Excess functions for the system 2-pentanol (1) + n-hexane (2) at $T = 313.15$ K: (—) G_m^E ; (---) H_m^E ; (- · - ·) TS_m^E . G_m^E was calculated using the Margules equation and H_m^E with the Redlich-Kister equation.

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List of symbols

- A_i : adjustable parameters of Redlich-Kister equation, Eq. (1)
 A_{ij}, A_{ji} : adjustable parameters of the VLE correlation models, Eqs. (2)–(5)
 B_{ii}, B_{ij}, B_{ji} : second virial coefficients
 C : adjustable parameter of Redlich-Kister equation, Eq. (1)
 G_m^E : excess molar Gibbs energy
 H_m^E : excess molar enthalpy
 i, j : constituent identification: 1 or 2
 $lit.$: literature value
 max : maximum value of the indicated quantity
 P : total pressure

P_i^s : vapour pressure of pure constituent i
 R : universal gas constant
 rms : root mean square
 S_m^E : excess molar entropy
 T : absolute temperature
 V_i^L : molar volume of pure liquid $i = 1, 2$
 x : mole fraction, liquid phase
 y : mole fraction, vapour phase

Greek letters

Δ : signifies difference
 α_{ij} : adjustable parameter in NRTL model, Eq. (4)
 $\lambda_{ij}, \lambda_{ji}$: adjustable parameters in six-parameter Margules equation, Eq. (2)
 η_{ij}, η_{ji} : adjustable parameters in six-parameter Margules equation, Eq. (2)